

ASSIGNMENT-4

Total Points: 100

Deadline: 11th April 2026

Instructions

For the assignments, please make sure to submit:

1. **PDF file** with code, outputs, and written explanations. Answer all questions in one single PDF.
2. **Code file** (.ipynb or .py). Do not submit zip files.

Question 1: Decision Trees and Dimensionality Reduction (50 Points)

For this question, you will use the **Breast Cancer Wisconsin (Diagnostic) dataset**. This dataset contains features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass, and the goal is to classify whether a tumor is malignant or benign.

You can load this directly using `sklearn.datasets.load_breast_cancer`.

Part A (25 Points)

1. Load the dataset and split it into a 70-30 train and test set.
2. Train a **Decision Tree Classifier** on the training data.
3. Calculate and report the performance of your model on the test set using Accuracy, F-1 score, Precision, and Recall.
4. Visualize the trained Decision Tree using `sklearn.tree.plot_tree` (you can limit the `max_depth` for a cleaner plot if necessary). Briefly explain the top root node—what feature was chosen for the first split and why?

Part B (25 Points)

1. Standardize your features using `StandardScaler`.
2. Apply **Principal Component Analysis (PCA)** to the standardized training data. Choose the number of components such that **90% of the variance** is retained. Apply the same transformation to your test set.
3. Train a *new* Decision Tree Classifier on this PCA-transformed data.
4. Calculate Accuracy, F-1 score, Precision, and Recall for this new model.
5. **Explain your findings:** Compare the performance metrics of the model in Part A with the model in Part B. Did the performance drop? How did dimensionality reduction affect the training process and the interpretability of the tree?

Question 2: A Comparative Study of Clustering Algorithms (50 Points)

In this question, you will use 2D synthetic datasets to visually understand the strengths and weaknesses of K-Means, Hierarchical Clustering, and DBSCAN.

Use the sklearn.datasets module to generate the following two datasets:

- **Dataset 1 (Blobs):** `make_blobs(n_samples=300, centers=4, cluster_std=0.60, random_state=0)`
- **Dataset 2 (Moons):** `make_moons(n_samples=300, noise=0.05, random_state=0)`

Part A: Finding the right "K" (15 Points)

1. Apply **K-Means Clustering** to **Dataset 1 (Blobs)**.
2. Use the **Elbow Method** (plotting the Within-Cluster Sum of Squares/Inertia against the number of clusters k from 1 to 10) to determine the optimal number of clusters.
3. Train a K-Means model with your chosen optimal k and create a scatter plot of the data points, coloring them by their cluster labels. Plot the cluster centroids clearly.

Part B: When Centroids Fail (15 Points)

1. Apply **K-Means Clustering** (with $k = 2$) to **Dataset 2 (Moons)**. Create a scatter plot of the results.
2. Apply **Agglomerative Hierarchical Clustering** (with `n_clusters=2` and `linkage='ward'`) to **Dataset 2 (Moons)**. Create a scatter plot of the results.
3. **Explain your findings:** Look at both plots. Did K-Means and Hierarchical clustering successfully group the two distinct moon shapes? Explain *why* these specific distance-based algorithms struggle with this specific data geometry.

Part C: Density-Based Clustering to the Rescue (20 Points)

1. Apply **DBSCAN** to **Dataset 2 (Moons)**.
2. You will need to tune the hyperparameters. Experiment with the `eps` (epsilon radius) and `min_samples` parameters until the algorithm perfectly separates the two moon shapes.
3. Create a scatter plot of the final DBSCAN results. (Make sure to distinguish noise points if any exist).
4. **Explain your findings:** What were your final `eps` and `min_samples` values? Explain conceptually why DBSCAN succeeds in clustering this dataset perfectly where K-Means and Hierarchical clustering failed.